

Query-Robust Chunk Summarization for Training-Free Sparse Attention

Literature Review, Robust Formulation, and Convex Solution

Technical note – September 2026

Abstract

Chunk-wise sparse attention reduces long-context decoding cost by replacing an exhaustive scan of the KV cache with a two-stage procedure: a lightweight selector first ranks pages or chunks, and exact attention is then evaluated only on the selected subset. Recent work has substantially improved the fidelity of the selector, progressing from key extrema and first-order mean summaries to dispersion corrections, second-order cumulants, page-local spectral reconstruction, and learned log-sum-exp surrogates. These developments primarily improve the *representation* used to score a chunk for a realized query. A distinct axis is the treatment of uncertainty in unseen or variable queries. Distributional methods optimize expected importance under a future-query model, whereas existing query-aware selectors are predominantly pointwise in the current query. This note formulates a robust alternative: construct a cacheable one-landmark chunk summary that minimizes the worst-case error in log attention mass over an explicit query uncertainty set. For an empirical convex-hull uncertainty set, the robust problem reduces exactly to a finite convex program over a probability simplex. Its dual is a smooth concave maximization whose solution recovers an optimal tangent of the chunk log-sum-exp. The optimal value is a minimax approximation radius that certifies uniform log-mass error over the uncertainty set.

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1. Literature Review and Query-Robustness Motivation

1.1. Chunk selection as attention-mass estimation

Consider a single attention head and a chunk C containing n keys $K_C = \{k_1, \dots, k_n\} \subset \mathbb{R}^d$. For a query $q \in \mathbb{R}^d$, define token logits

$$s_j(q) = \frac{q^\top k_j}{\sqrt{d}}, \quad (1)$$

the unnormalized chunk attention mass

$$Z_C(q) = \sum_{j \in C} \exp(s_j(q)), \quad (2)$$

and the log-mass

$$f_C(q) = \log Z_C(q) = \log \sum_{j \in C} \exp\left(\frac{q^\top k_j}{\sqrt{d}}\right). \quad (3)$$

For a fixed query, ranking chunks by normalized attention mass, by $Z_C(q)$, or by $f_C(q)$ is equivalent because the global softmax denominator is common to all chunks and the logarithm is monotone. HiLS formalizes exact block-sparse selection in precisely this form, and COBS independently derives block attention mass as the value-agnostic selection criterion under explicit assumptions [6, 4]. LOCKS adopts exact page-level log-sum-exp mass as the selection target and reconstructs sufficient within-page logits to estimate it directly [5].

This observation separates two issues:

1. **Selection target:** which scalar should define an ideal block ranking? For value-blind deterministic routing, recent analyses identify attention mass as a principled target [6, 4, 5].
2. **Mass estimator:** how can $f_C(q)$ be estimated from a cacheable summary without reading all keys in C ?

The second issue is the central design axis of query-aware chunk selectors.

1.2. Progression of cacheable chunk summaries

Quest: extrema and a max-logit envelope. Quest stores per-dimension minima and maxima of keys in each page. Given the current query, it computes a page score that upper-bounds the largest token logit attainable in the page and retrieves the top-scoring pages [1]. Its primitive is therefore closer to an envelope of

$$\max_{j \in C} s_j(q) \quad (4)$$

than to a direct approximation of the entire log-mass in (3). Since

$$\max_j s_j(q) \leq f_C(q) \leq \max_j s_j(q) + \log n, \quad (5)$$

this proxy is particularly appropriate in a dominant-token regime, but it need not preserve the ordering induced by total chunk mass when many moderately large logits contribute jointly.

ShadowKV: first-order mean landmark with explicit outlier handling. ShadowKV uses compact key representations and a sparse KV selection mechanism within a system that combines low-rank key storage with value offloading [2]. A central nonparametric chunk summary is the mean key. For equal-size chunks,

$$\bar{k}_C = \frac{1}{n} \sum_{j=1}^n k_j, \quad \frac{q^\top \bar{k}_C}{\sqrt{d}} = \frac{1}{n} \sum_j s_j(q). \quad (6)$$

Mean-based scoring is exact at first order around zero query, but it suppresses within-chunk variation. ShadowKV therefore includes explicit treatment for chunks poorly represented by the mean. This exposes a recurring failure mode: a small number of keys can be functionally important for particular query directions while contributing weakly to a raw average.

UNIQUE: first moment plus scalar dispersion. UNIQUE augments a page mean with a scalar standard-deviation correction,

$$\text{score}_C(q) = q^\top \bar{k}_C + \lambda \|q\|_2 \text{std}_C, \quad (7)$$

where the dispersion term is intended to compensate for salient keys diluted by mean pooling [3]. This is training-free and cacheable, but the correction is isotropic: it does not identify which directions of the query space induce the large within-page variation.

COBS: second-order cumulant information. COBS explicitly expands the block log-mass as the cumulant generating function of the empirical key distribution. With mean $\bar{k}_C$ and covariance Σ_C ,

$$f_C(q) = \log n + q^\top \bar{k}_C + \frac{1}{2} q^\top \Sigma_C q + \text{higher-order cumulants}, \quad (8)$$

where the attention scaling can be absorbed into q for notational simplicity. COBS stores a compressed second-order statistic and uses the quadratic term to restore curvature omitted by first-order selectors [4]. This directly improves *representation order*: mean-only summaries are first-order; COBS retains a compressed second-order term.

LOCKS: page-local spectral reconstruction. LOCKS takes a higher-capacity route. Each page stores its own low-rank spectral summary. At decode time, the method reconstructs approximate within-page logits and computes their log-sum-exp, thereby estimating the established mass target directly without reading candidate keys or values [5]. LOCKS is training-free and demonstrates that page-local structure can be materially more faithful than a shared or sequence-global low-rank representation.

HiLS: learned affine log-mass surrogate. HiLS starts from the exact chunk mass and derives an affine surrogate for the log-sum-exp. Given a learnable land-

mark query q'_C , it defines

$$p_j = \text{softmax}_j \left(\frac{q'_C{}^\top k_j}{\sqrt{d}} \right), \quad (9)$$

$$k'_C = \sum_j p_j k_j, \quad (10)$$

$$b'_C = - \sum_j p_j \log p_j, \quad (11)$$

and scores the chunk with

$$\hat{f}_C(q) = \frac{q^\top k'_C}{\sqrt{d}} + b'_C. \quad (12)$$

The summary is integrated into a hierarchical attention mechanism and optimized end-to-end under language-modeling loss; conversion of dense models requires continued training [6]. The functional form in (12) is especially relevant because it is exactly the tangent form of the log-sum-exp, as shown in Section 2.

1.3. A second axis: how query uncertainty is represented

The preceding sequence primarily increases *summary richness*. A different question is how a method treats variability or uncertainty in the query.

Expected Attention is the clearest example of explicit future-query modeling. It assumes hidden states, and hence projected queries, are approximately Gaussian and computes expected unnormalized attention analytically using the Gaussian moment generating function [7]. This is an *average-case* criterion. In contrast, Quest, ShadowKV, UNIQUE, COBS, and LOCKS primarily evaluate a cacheable summary at the realized current query. Q-Filters and RetrievalAttention demonstrate that query geometry can be exploited directly and that query and key distributions need not be interchangeable [8, 9].

These works suggest a two-dimensional research map:

Representation richness: extrema/mean $\rightarrow$ dispersion $\rightarrow$ covariance $\rightarrow$ local spectral reconstruction. Query criterion: pointwise $\rightarrow$ expectation under P_Q $\rightarrow$ worst case over $\mathcal{U}_Q$.
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Recent chunk selectors have advanced substantially along the first axis. The second axis remains less developed for cacheable one-landmark summaries.

1.4. Motivation: robust approximation under query uncertainty

For an unseen or variable query, minimizing an expected error

$$\mathbb{E}_{q \sim P_Q} [e_C(q)] \quad (13)$$

can assign negligible weight to rare queries even when they induce catastrophic ranking errors. This distinction is important when a chunk contains a *functional outlier*: a key that dominates attention only for a small region of query space. Mean summaries may dilute such a key, and expectation-based criteria may tolerate the associated error if the responsible queries have low probability.

Table 1: Representative methods organized by summary target and treatment of queries. “Training-free” denotes no update of model parameters; calibration may still be used.

Method	Cached/selection summary	Query treatment	Training-free	Relation to query robustness
Quest [1]	Per-page key minima/maxima	Current query is observed exactly	Yes	Pointwise query-aware max-logit envelope
ShadowKV [2]	Mean-like chunk landmark plus outlier handling; low-rank key representation	Current query is observed exactly	Yes	Pointwise scoring; no explicit uncertainty set
UNIQUE [3]	Mean + scalar key dispersion	Current query is observed exactly	Yes	Pointwise scoring with query-norm-dependent dispersion correction
COBS [4]	Mean + compressed covariance	Current query; optional query subspace statistics for compression	Not a pure drop-in TF method	Improves approximation order, not worst-case query coverage
LOCKS [5]	Page-local low-rank spectral factor	Current query is observed exactly	Yes	High-fidelity pointwise LSE reconstruction
HiLS [6]	Learned weighted key + entropy bias	Landmark and query calibration learned by LM objective	No	Learns a surrogate; no explicit worst-case query set
Expected Attention [7]	Per-token expected contribution	Future query modeled as Gaussian	Yes	Distributional, expectation-based future-query modeling
Q-Filters [8]	Context-agnostic QK projection	Empirical query geometry summarized offline	Yes	Uses query statistics, but not a minimax uncertainty objective
Retrieval-Attention [9]	Query-aware ANN index	Explicitly addresses Q/K distribution mismatch	Yes	Evidence against treating key geometry as a direct query model

A robust alternative is to define a compact uncertainty set $\mathcal{U}_Q$ of plausible queries and optimize

$$\sup_{q \in \mathcal{U}_Q} e_C(q). \quad (14)$$

Among the reviewed works, no method was identified that constructs a cacheable one-landmark chunk summary by directly minimizing the worst-case log-mass approximation error over an explicit query uncertainty set. This motivates the formulation below.

2. Query-Robust Formulation and Solution

2.1. Exact target and the one-landmark surrogate family

Fix a chunk C and write $K \in \mathbb{R}^{n \times d}$ for the matrix whose rows are $k_j^\top$. Define $f(q)$ as in (3). The Gibbs variational identity gives

$$f(q) = \max_{p \in \Delta^{n-1}} \left\{ \frac{q^\top K^\top p}{\sqrt{d}} + H(p) \right\}, \quad (15)$$

where

$$H(p) = - \sum_{j=1}^n p_j \log p_j. \quad (16)$$

Consequently, every fixed $p \in \Delta^{n-1}$ induces a cacheable affine lower bound

$$T_p(q) = \frac{q^\top \bar{k}(p)}{\sqrt{d}} + H(p), \quad \bar{k}(p) = K^\top p, \quad (17)$$

with

$$T_p(q) \leq f(q) \quad \forall q. \quad (18)$$

Thus a chunk can be represented at runtime by one vector and one scalar, $(\bar{k}(p), H(p))$.

Proposition 1 (Mean pooling is the uniform member of the surrogate family). *For $p = u = (1/n)\mathbf{1}$,*

$$T_u(q) = \log n + \frac{q^\top \bar{k}}{\sqrt{d}}, \quad \bar{k} = \frac{1}{n} \sum_j k_j. \quad (19)$$

Hence the standard mean-key score, up to the constant $\log n$ for equal-size chunks, is a special case of (17).

This representation family therefore contains the raw mean but permits a nonuniform weighting of keys, allowing a rare key to receive disproportionate mass if required by robustness.

2.2. Robust objective

Let $\mathcal{U}_Q \subset \mathbb{R}^d$ denote the query uncertainty set for a layer/head. Define the one-landmark robust problem

$$\epsilon_C^* = \min_{p \in \Delta^{n-1}} \sup_{q \in \mathcal{U}_Q} [f_C(q) - T_{C,p}(q)]. \quad (20)$$

The optimizer p_C^* gives the robust representative

$$\bar{k}_C^* = K_C^\top p_C^*, \quad b_C^* = H(p_C^*), \quad (21)$$

and ϵ_C^* is the minimax approximation radius.

The preferred training-free uncertainty model is an empirical polytope constructed from frozen-model query activations,

$$\mathcal{U}_Q = \text{conv}\{q_1, \dots, q_M\}. \quad (22)$$

The q_i may be collected from a small calibration pass or from observed queries of the current sequence. No model parameter is optimized. This is training-free, although the calibration-based variant is not calibration-free. The use of actual query activations avoids assuming that queries lie in the convex hull of keys; RetrievalAttention provides empirical motivation for maintaining this distinction because query and key distributions can exhibit substantial mismatch [9].

2.3. KL and Bregman interpretations

Define the within-chunk attention distribution

$$a(q) = \text{softmax}\left(\frac{Kq}{\sqrt{d}}\right) \in \Delta^{n-1}. \quad (23)$$

Proposition 2 (Exact error identity). *For every $p \in \Delta^{n-1}$ and every query q ,*

$$\boxed{f(q) - T_p(q) = D_{\text{KL}}(p \| a(q))}. \quad (24)$$

Proof. Since $\log a_j(q) = q^\top k_j / \sqrt{d} - f(q)$,

$$\begin{aligned} D_{\text{KL}}(p \| a(q)) &= \sum_j p_j \log p_j - \sum_j p_j \log a_j(q) \\ &= -H(p) - \frac{q^\top K^\top p}{\sqrt{d}} + f(q) \\ &= f(q) - T_p(q). \end{aligned}$$

□

Therefore (20) is equivalent to the information-radius problem

$$\boxed{p_C^* = \arg \min_{p \in \Delta^{n-1}} \sup_{q \in \mathcal{U}_Q} D_{\text{KL}}(p \| a_C(q))}. \quad (25)$$

The robust representative is the probability vector that minimizes its largest reverse-KL distance to any attention response induced by a plausible query. This formalizes *query-response heterogeneity*: a chunk is difficult to represent by one landmark when plausible queries induce substantially different within-chunk attention distributions.

When $p = a(q_0)$ for some anchor q_0 , T_p is exactly the first-order tangent of f at q_0 and

$$f(q) - T_{a(q_0)}(q) = D_f(q, q_0), \quad (26)$$

where D_f is the Bregman divergence generated by f . Hence the term *minimax tangent* is exact whenever the robust optimizer is realizable as $a(q_0)$; Section 2.5 proves that this holds for the empirical convex-hull uncertainty set.

2.4. Finite reduction for a convex-hull query set

For fixed p , the error

$$g_p(q) = f(q) - T_p(q) \quad (27)$$

is convex in q because f is convex and T_p is affine.

Proposition 3 (Vertex reduction). *Let $\mathcal{U}_Q = \text{conv}\{q_1, \dots, q_M\}$. Then*

$$\sup_{q \in \mathcal{U}_Q} g_p(q) = \max_{i=1, \dots, M} g_p(q_i). \quad (28)$$

Thus the continuous robust problem reduces exactly to

$$\epsilon_C^* = \min_{p \in \Delta^{n-1}} \max_{i \leq M} D_{\text{KL}}(p \| a_i), \quad a_i = a(q_i). \quad (29)$$

Equivalently, using an epigraph variable t ,

$$\begin{aligned} \min_{p, t} \quad & t \\ \text{s.t.} \quad & D_{\text{KL}}(p \| a_i) \leq t, \quad i = 1, \dots, M, \\ & p \in \Delta^{n-1}. \end{aligned} \quad (30)$$

Because $D_{\text{KL}}(p \| a_i)$ is convex in its first argument, (30) is a finite-dimensional convex optimization problem. The softmax distributions a_i are strictly positive, so standard interior-point, mirror-descent, or first-order convex solvers can compute the global optimum.

2.5. Dual formulation and recovery of the optimal tangent

The convex program admits a compact dual. Introduce $\lambda \in \Delta^{M-1}$ as multipliers for the M KL constraints.

Theorem 1 (Dual minimax solution). *For $\mathcal{U}_Q = \text{conv}\{q_1, \dots, q_M\}$, strong duality holds and*

$$\epsilon_C^* = \max_{\lambda \in \Delta^{M-1}} \left[\sum_{i=1}^M \lambda_i f_C(q_i) - f_C \left(\sum_{i=1}^M \lambda_i q_i \right) \right]. \quad (31)$$

If λ^* solves (31), define

$$q_C^* = \sum_i \lambda_i^* q_i. \quad (32)$$

Then the primal optimum is

$$p_C^* = a_C(q_C^*) = \text{softmax} \left(\frac{K_C q_C^*}{\sqrt{d}} \right). \quad (33)$$

Consequently the minimax-optimal affine lower bound is a genuine tangent of f_C at the anchor $q_C^* \in \mathcal{U}_Q$.

Proof sketch. The Lagrangian of (30) enforces $\sum_i \lambda_i = 1$. Minimizing the weighted KL objective over $p \in \Delta^{n-1}$ yields

$$p_j^*(\lambda) \propto \prod_i a_{i,j}^{\lambda_i}. \quad (34)$$

Since $a_i = \text{softmax}(K q_i / \sqrt{d})$, this geometric mixture equals

$$p^*(\lambda) = \text{softmax} \left(\frac{K \bar{q}(\lambda)}{\sqrt{d}} \right), \quad \bar{q}(\lambda) = \sum_i \lambda_i q_i. \quad (35)$$

Substitution gives the dual objective

$$\sum_i \lambda_i f(q_i) - f(\bar{q}(\lambda)), \quad (36)$$

which is concave in λ because f is convex. Slater's condition holds, yielding strong duality. $\square$

Equation (31) has an additional interpretation: ϵ_C^* is the maximum Jensen gap of the convex function f_C over mixtures of the uncertainty-set vertices. The dual therefore chooses a mixture of adversarial queries whose dispersion induces the largest nonlinearity of the chunk log-mass; the primal places the tangent at their convex combination.

2.6. Direct numerical solution

For a chunk of size n and M query vertices, first compute the $M \times n$ logit matrix

$$S_{ij} = \frac{q_i^\top k_j}{\sqrt{d}}, \quad (37)$$

and vertex log-masses

$$f_i = \log \sum_j e^{S_{ij}}. \quad (38)$$

The dual objective is

$$\phi(\lambda) = \sum_i \lambda_i f_i - f(Q^\top \lambda), \quad \lambda \in \Delta^{M-1}, \quad (39)$$

where Q has rows $q_i^\top$. Its gradient is

$$\frac{\partial \phi}{\partial \lambda_i} = f_i - q_i^\top \nabla f(\bar{q}), \quad \bar{q} = \sum_i \lambda_i q_i. \quad (40)$$

A projected-gradient or exponentiated-gradient ascent on the simplex therefore yields the global optimum of the smooth concave dual. For typical page sizes and a small set of query vertices, this is a small offline or incremental preprocessing problem; runtime routing retains only one dot product and one scalar bias per chunk.

A minimal algorithm is:

1. Collect frozen-model queries $\{q_i\}_{i=1}^M$ for a given layer/head and define $\mathcal{U}_Q$ by (22).
2. For each chunk, compute $f_C(q_i)$ and solve the concave dual (31) for λ_C^* .
3. Recover q_C^* by (32), then p_C^* by (33).

4. Store

$$\bar{k}_C^* = K_C^\top p_C^*, \quad b_C^* = H(p_C^*), \quad \epsilon_C^* = \phi(\lambda_C^*). \quad (41)$$

5. At decoding time, score the chunk using

$$\hat{f}_C(q) = \frac{q^\top \bar{k}_C^*}{\sqrt{d}} + b_C^*. \quad (42)$$

2.7. Uniform certificate and relation to routing

By construction,

$$0 \leq f_C(q) - \hat{f}_C(q) \leq \epsilon_C^*, \quad \forall q \in \mathcal{U}_Q. \quad (43)$$

Thus each chunk has a certified interval

$$\hat{f}_C(q) \leq f_C(q) \leq \hat{f}_C(q) + \epsilon_C^*. \quad (44)$$

For two chunks A and B , if

$$\hat{f}_A(q) + \epsilon_A^* < \hat{f}_B(q), \quad (45)$$

then the true ordering is certified as $f_A(q) < f_B(q)$. More generally, if a symmetric estimator is desired, define

$$\tilde{f}_C(q) = \hat{f}_C(q) + \frac{\epsilon_C^*}{2}, \quad (46)$$

which satisfies

$$|f_C(q) - \tilde{f}_C(q)| \leq \frac{\epsilon_C^*}{2} \quad \forall q \in \mathcal{U}_Q. \quad (47)$$

Hence, if the true log-mass gap between the K -th and $(K+1)$ -th chunks exceeds the sum of the relevant approximation radii, the top- K ordering is preserved.

2.8. Interpretation of the minimax radius

The optimal value

$$\epsilon_C^* = \min_p \max_{q \in \mathcal{U}_Q} D_{\text{KL}}(p \| a_C(q)) \quad (48)$$

measures the smallest unavoidable worst-case error when the entire chunk is compressed to one affine landmark. It should therefore be interpreted primarily as a *single-landmark compression radius* or *query-response heterogeneity* measure, rather than as a geometric outlier score.

A large radius may arise from one rare extreme key, several competing clusters, or any other structure causing plausible queries to induce substantially different attention distributions. This distinction is stronger than raw key-space dispersion: a geometrically distant key matters only if some query direction in $\mathcal{U}_Q$ makes it functionally relevant.

Corollary 1 (Worst-case dominance over raw mean). *Let $u = (1/n)\mathbf{1}$. Since $u \in \Delta^{n-1}$ is feasible in (20),*

$$\epsilon_C^* \leq \sup_{q \in \mathcal{U}_Q} [f_C(q) - T_u(q)]. \quad (49)$$

Thus the robust one-landmark summary cannot have larger worst-case log-mass error than the raw mean member of the same surrogate family.

2.9. Scope and research hypothesis

The formulation isolates a specific hypothesis:

At a fixed one-landmark summary budget, chunk quality is governed by the diversity of attention responses induced by plausible queries, rather than by key-space dispersion alone. Minimizing worst-case log-mass error over a frozen-model query uncertainty set should therefore improve robust chunk ranking relative to a raw mean landmark, especially for functionally heterogeneous chunks.

This hypothesis is deliberately narrower than claims of universally better mass approximation. COBS and LOCKS increase representation capacity through second-order or local spectral summaries; HiLS learns an expressive affine surrogate end-to-end. The proposed direction instead fixes an extremely small representation budget and changes the optimization criterion from pointwise or expected query behavior to worst-case query coverage.

Conclusion

The current sparse-attention literature increasingly converges on attention mass as the appropriate value-blind chunk-selection target. The main line of progress has enriched the cacheable summary used to estimate this mass: extrema, means, dispersion terms, covariances, spectral reconstructions, and learned affine surrogates. Query robustness provides an orthogonal axis. For an empirical convex-hull uncertainty set built from frozen-model queries, the one-landmark robust problem is mathematically tractable: the worst case reduces to the hull vertices, the primal is convex in the representative distribution, the dual is a smooth concave Jensen-gap maximization, and the optimizer recovers an exact tangent of the chunk log-sum-exp. The associated minimax radius supplies a uniform approximation certificate and an intrinsic measure of single-landmark compressibility.

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